

Countervailing Market Power and Consumer Surplus: An Empirical Examination

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Vertical markets present several challenges for antitrust. Of particular relevance is the question of how to deal with horizontal agreements that seek to counteract a large imbalance of power in vertical negotiations. Although these agreements are normally considered antitrust offenses, their prosecution has recently been called into question by antitrust practitioners on the grounds that bargaining should be more balanced (see, e.g., Mullan and Timan, 2018; ACCC, 2018). There are many examples of negotiations with potentially large imbalances in bargaining positions, including the negotiations between physicians and insurance companies, between hotels and large lodging platforms, and between bus owners and transportation companies. And yet, there is almost no empirical evidence analyzing the consequences for consumers of these types of agreements.

We focus on a case of physicians in negotiations with insurers. Collective negotiation of physicians with payers is a recurrent and commonly cited case of horizontal agreement in the context of vertical negotiations, and the leading example cited in much of the academic debate (see, e.g., Blair and Herndon, 2004; Blair and DePasquale, 2011; Campbell, 2007, 2008; Baker, Farrell and Shapiro, 2008). The high degree of con-

centration in the health insurance market is a likely reason for their occurrence, as well as for the emergence of proposals seeking to counteract insurer bargaining power (see Fulton, 2017 and American Medical Association, 2011). The FTC has generally deemed such agreements between physicians anti-competitive. However, Alaska, New Jersey, and Texas have passed bills that permit physicians to bargain collectively (Blair and Herndon, 2004). More recently, the DOJ and FTC have explicitly allowed joint negotiations of providers in the context of the recently formed “Accountable Care Organizations” (DOJ and FTC, 2011, p. 67028).

This paper contributes to the literature by empirically assessing the consequences for patients of a recent case involving physicians in their negotiations with payers. In Alé-Chilet and Atal (2020), we analyze the same case in detail, and provide an analysis of the strategies employed by the Association and its members. This article complements our main study by presenting estimates of the impact of the physicians’ actions on patients’ out-of-pocket costs and consumer surplus.

I. Institutions

We provide a brief summary of the relevant institutional details (see Alé-Chilet and Atal, 2020, for more details). The focus of this paper is on the private arm of the Chilean health care system, which is operated by a group of six private insurance companies known collectively as Isapres (“Instituciones de Salud Previsional”). Isapres cover around 17 percent of the population. The private system exists alongside a public option, FONASA, which covers roughly two thirds of the population

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(about 11 million people).¹

Plans in the private system are comprehensive, stand-alone plans. These plans have two main coverage features: coinsurance rates (one for inpatient care and another for outpatient care) and coverage caps (insurer payment caps). The insurance plans in our data have a coverage structure that is similar to “preferred provider organization” (PPO) plans. Plans provide higher coverage for visits to a list of specified “in-network” providers than to “out-of-network” providers.

Health care provision is also divided between public and private providers. As is common in private health care markets, Isapres pay their in-network providers a negotiated rate, while out-of-network providers set their rates to Isapres unilaterally. Finally, FONASA reimburses private providers following a pre-determined rate for each procedure and provider quality tier.

II. The Case

The focus of our study is a recent antitrust case against a trade association of OB/GYNs and its members. In an effort to improve their bargaining position vis-à-vis the insurance companies, 26 of the 29 OB/GYNs in the city of Chillán formed a trade association and started to jointly negotiate their fees with Isapres in early 2011. After unsuccessful negotiations, in December 2011 the Association agreed to undertake three measures: First, all members would jointly cancel their contracts with Isapres and become “out-of-network” providers. Second, members would only charge fees above a commonly-agreed-upon minimum level, that was almost 80 percent higher than the average fee in 2011. Finally, all future negotiations between the members and the insurers would be carried out through a designated representative. These

measures went into effect in February 2012.

Throughout the rest of 2012, the insurance companies and the doctors engaged in unsuccessful negotiations. In March 2013 the Fiscalía Nacional Económica (FNE), or National Economic Prosecutor, started investigating the Association for antitrust offenses, and filed an indictment in October 2013. The Competition Tribunal found the Association members guilty of antitrust offenses in 2015.

III. Data

We use a combination of administrative data from the insurance companies and receipts collected by the FNE to construct the universe of visits from privately-insured patients to the gynecologists operating in Chillán and its neighboring provinces. For each visit between January 2009 and March 2013, we observe the date, physician and patient identifiers, price, and out-of-pocket expenditures. We also obtain the physician’s main address from the case documents (FNE, 2013) and local websites. After dropping physicians with very few visits, our final sample includes 25 associated and two non-associated physicians.

Figure 1 shows the evolution of average out-of-pocket costs—which combine physicians’ list prices with coinsurance rates—for Chillán and its two largest neighboring cities. As the graph indicates, out-of-pocket costs rose sharply in Chillán after the measures of the Association went into effect. Key to our empirical strategy, out-of-pocket costs in neighboring cities had a similar level and followed a similar trend up to the point when the measures of the Association in Chillán went into effect.

IV. Empirical Analysis

We formalize the graphical result from Figure 1 with the estimation of a difference-in-differences model to quantify the effect of the Association’s measures on out-of-pocket costs. Specifically, we estimate by ordinary least squares (OLS) the following specification for average out-of-pocket costs $\overline{oop}_{dt}$ in municipality d and month t :

¹A small fraction of the population is insured by seven “closed” private insurance companies, which are available only to workers in certain industries. The rest are covered by a special system, such as those in the Armed Forces—or do not have coverage at all (Bitran, Escobar and Gassibe (2010)).

$$(1) \overline{oop}_{dt} = \alpha_t + T_d + \beta \times T_d \times \mathbb{1}[t \geq t_0] + \epsilon_{dt},$$

where $T_d \equiv \mathbb{1}[d = \text{Chillán}]$ is an indicator for Chillán—the city where the Association operated—and $\mathbb{1}[t \geq t_0]$ is an indicator variable that takes the value of 1 for periods after the Association’s measure went into effect—i.e., every month starting from February 2012. The parameter β represents the causal effect of the Association’s strategies on average out-of-pocket costs.

Table 1 presents the estimation results. The table indicates that $\hat{\beta}$ is CLP 6,090 with a standard error of CLP 358. Considering that the average out-of-pocket cost in Chillán before February 2012 was equal to CLP 3,770, this effect corresponds to a 160 percent increase in out-of-pocket costs for patients of the average physician in Chillán.

Next, we use the demand model for physicians in Alé-Chilet and Atal (2020) to calculate the impact of the physician trade association on consumer surplus. In the model, each combination of insurer and month corresponds to a different market. Every month, patients of each insurer may choose from among any of the physicians in Chillán, a physician in a neighboring city, or the outside option. The mean patient utility of a visit to physician i for a patient insured with Isapre j at time t is

$$\delta_{ijt} = \alpha oop_{ijt} + f_{ijt},$$

where α is the marginal utility of income, oop_{ijt} is the average out-of-pocket expenses for patients insured with j who visit physician in month t , and f_{ijt} represents a collection of fixed effects that includes physician-insurer fixed effects, month-of-the-year fixed effects to control for seasonality, and year fixed effects to control for common market trends. The random disturbance around the mean utility has a nested logit structure, in which groups of physicians, or “nests,” are determined according to each doctor’s geographic location and are aimed to capture a patient’s distaste for travel distance—a well-documented feature of the demand for health care services (see,

e.g., Phelps and Newhouse, 1974).

We estimate the demand model using methods developed by Berry (1994) and Gandhi, Lu and Shi (2014). The latter allows us to deal with the absence of private patients for roughly a fourth of the doctor-insurer-month combinations. To identify the parameters, we leverage the sharp increase in out-of-pocket costs brought about by the Association as a source of exogenous price variation. The demand estimates are provided in Alé-Chilet and Atal (2020).

Let $D_{jkt} = \sum_{i \in k} \exp\left(\frac{\delta_{ijt}}{1-\sigma}\right)$, where k denotes a nest, and σ is the within-nest correlation parameter estimated in the demand model. Given our parametric assumptions for utility, the expected total consumer surplus in a given month t takes the usual form (see, e.g., Anderson, De Palma and Thisse, 1992):

$$(2) \quad CS_t = \frac{1}{\alpha} \sum_j M_j \ln \left(1 + \sum_k D_{jkt}^{1-\sigma} \right) + A,$$

where M_j is the number of female insurees in Isapre j (the market size), and A is a constant. This constant drops out when we consider changes in consumer surplus.

We use Equation (2) to calculate CS_t for every month in the year after February 2012, under both the observed prices and the counterfactual scenario in which doctors had kept their prices at their pre-February 2012 average. We find that the trade association’s measures caused an average decrease of CLP 4.7 million (USD 9,381) in consumer surplus each month after February 2012, which amounts to a decrease in surplus of CLP 93.8 million in the 20 months until the indictment was filed. This drop in consumer surplus corresponds to 151 percent of total patient spending in the pre-February 2012 period and to 43 percent of combined physician revenues in the same period.

V. Conclusion

In this paper we contribute to the discussion regarding whether to allow hori-

zontal agreements that seek to counteract large imbalances of power in vertical negotiations. We show how detailed data can be used to analyze the effects of such a case in the context of physicians negotiation with insurers.

Our findings show large increases in out-of-pocket costs with corresponding large decreases in patient surplus in the case we analyze. These findings are a direct consequence of the breakup of a vertical relationship that restrained physicians' rates substantially. Although our results are not immediately generalizable to other settings, they show that antitrust authorities should exercise caution in deciding whether to approve these types of agreements.

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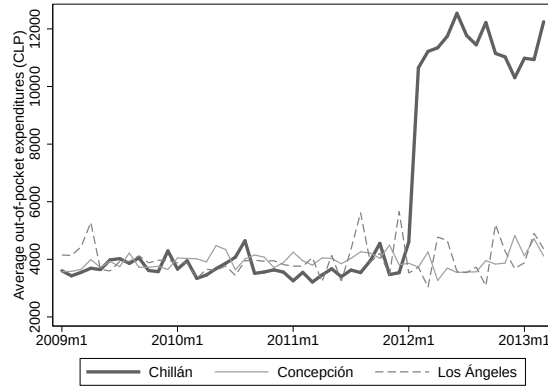


FIGURE 1. AVERAGE OUT-OF-POCKET EXPENDITURES

Note: The Figure shows patients' median out-of-pocket expenditures at the average gynecologist, in Chillán and in the two major neighboring cities of Concepción and Los Ángeles. Prices are in Chilean Pesos (approx. CLP 500 = USD 1).

TABLE 1—DIFFERENCE-IN-DIFFERENCES ESTIMATES OF OUT-OF-POCKET EXPENDITURES

	$\overline{oop}$
T_d	-541.66*** (106.34)
$1[t \geq t_0]$	2,186.77*** (473.92)
$T_d \times 1[t \geq t_0]$	6,090.79*** (358.11)
$\text{mean}(\overline{oop})$	3,770.26
N	5,813
R^2	0.14

Robust standard errors in parentheses

* p<0.10, ** p<0.05, *** p<0.01

Note: The Table shows the coefficient estimates of Equation (1)—i.e., median out-of-pocket expenditures, and the change in it, faced by patients of the average gynecologist in Chillán relative to patients in two major neighboring cities (Concepción and Los Ángeles). Out-of-pocket expenditures are measured in Chilean Pesos (approx. CLP 500 = USD 1).